

Lecture 22 — Kernel Modules and eBPF — Extending the Kernel

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Placeholder

This lecture has not yet been written. The detailed outline lives in the plan file `let-s-look-at-containers-structured-sprout.m`.

The intended sections are:

1. Why extend the kernel?
2. Loadable Kernel Modules — anatomy
3. Why LKMs are dangerous
4. Module trust, signing, and the BYOVD problem
5. eBPF as the modern alternative
6. The eBPF verifier
7. Maps and the kernel/user boundary
8. Where eBPF is going
9. Monolithic vs. microkernel revisited

References